

VOROGOVO, Russian Federation

WMO: 239730

Lat: 61.031N

Lon: 89.614E

Elev: 49

StdP: 100.74

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-43.5	-40.3	-45.1	0.0	-42.5	-42.6	0.1	-39.9	11.1	-11.5	9.9	-16.3	1.2	0	0.682

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.1	28.3	18.7	26.4	18.0	24.5	17.4	20.4	25.5	19.4	24.1	18.3	22.7	2.8	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.7	13.6	22.7	17.6	12.7	21.8	16.6	11.9	20.7	59.3	25.8	55.8	24.2	52.3	22.6	26.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
9.1	7.9	7.0	-44.7	31.3	3.8	1.6	-47.4	32.4	-49.7	33.4	-51.8	34.3	-54.6	35.5		
			-43.7	22.2	2.8	1.5	-45.8	23.3	-47.4	24.2	-49.0	25.0	-51.0	26.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	-2.3	-23.4	-19.2	-9.8	-1.7	6.0	15.4	18.3	14.5	7.6	-1.3	-14.0	-21.1
	DBStd	15.90	9.97	8.51	7.11	5.51	5.22	4.90	3.54	3.85	4.24	5.80	9.43	10.00
	HDD10.0	5114	1036	818	613	352	147	12	0	6	97	352	719	963
	HDD18.3	7609	1294	1051	871	602	382	112	45	129	323	610	969	1221
	CDD10.0	627	0	0	0	0	24	174	258	146	24	1	0	0
	CDD18.3	81	0	0	0	0	1	24	44	11	0	0	0	0
	CDH23.3	644	0	0	0	0	8	234	326	76	1	0	0	0
	CDH26.7	138	0	0	0	0	1	53	71	14	0	0	0	0
Wind	WSAvg	3.0	3.2	3.0	3.2	3.2	3.2	2.7	2.5	2.6	2.9	3.3	3.0	3.2
Precipitation	PrecAvg	565	33	28	25	28	46	66	59	66	64	62	50	41
	PrecMax	782	70	49	51	58	108	142	106	114	109	100	91	83
	PrecMin	413	1	9	7	11	14	22	7	13	23	36	19	17
	PrecStd	82	16	9	11	12	22	28	28	26	24	15	17	17
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-1.4	-0.5	5.5	13.4	24.2	30.5	31.0	28.6	21.2	12.2	2.2	-0.8
		MCWB	-1.8	-1.0	2.0	6.6	14.8	18.3	20.4	20.0	14.6	8.9	1.4	-1.2
	2%	DB	-4.3	-2.9	3.2	10.3	20.5	28.3	28.5	25.5	18.2	9.7	1.2	-2.9
		MCWB	-4.6	-3.7	0.4	5.1	12.4	17.6	19.5	18.3	13.4	7.7	0.6	-3.4
	5%	DB	-7.2	-5.2	1.3	7.8	17.6	26.2	26.7	23.1	16.0	7.7	0.3	-4.9
		MCWB	-7.6	-5.8	-0.5	3.8	11.0	17.1	18.7	17.2	12.4	6.0	-0.2	-5.4
	10%	DB	-9.9	-7.7	-0.1	5.6	14.5	24.0	25.0	20.8	13.9	5.8	-1.1	-7.4
		MCWB	-10.3	-8.3	-1.5	2.4	9.2	16.4	18.2	16.4	11.2	4.4	-1.7	-7.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-1.7	-0.8	2.5	7.7	15.4	20.5	22.4	20.9	15.6	9.8	1.6	-1.2
		MCDB	-1.4	-0.4	4.7	11.6	22.5	26.6	27.8	26.2	19.0	11.5	1.8	-1.0
	2%	WB	-4.6	-3.5	0.8	5.7	13.2	19.2	21.0	19.4	14.1	7.8	0.8	-3.4
		MCDB	-4.3	-3.0	2.5	9.7	19.1	24.7	25.9	23.9	17.3	9.4	1.1	-3.0
	5%	WB	-7.5	-5.8	-0.3	4.1	11.5	18.2	20.1	18.0	12.8	6.3	-0.2	-5.3
		MCDB	-7.2	-5.2	1.2	7.1	16.6	23.5	25.0	21.7	15.4	7.6	0.3	-5.0
	10%	WB	-10.3	-8.2	-1.6	2.8	9.8	17.2	19.1	16.7	11.5	4.5	-1.6	-7.9
		MCDB	-9.9	-7.7	-0.2	5.2	14.4	22.7	23.3	20.2	13.7	5.6	-1.1	-7.5
Mean Daily Temperature Range	5% DB	MDBR	7.2	8.6	10.2	9.2	8.9	9.5	9.1	7.9	6.5	4.8	6.4	7.3
		MCDBR	7.3	8.9	9.2	11.4	13.3	13.9	12.2	11.5	10.1	6.1	4.7	8.3
		MCWBR	7.3	8.8	7.5	7.4	7.0	5.5	5.1	5.5	5.7	4.6	4.8	8.3
	5% WB	MCDBR	7.6	9.4	8.1	9.7	11.8	10.9	10.3	10.0	8.4	5.8	5.0	8.4
		MCWBR	7.6	9.2	7.0	6.9	6.7	5.2	4.9	5.5	5.3	4.6	5.1	8.3
Clear-Sky Solar Irradiance	taub	0.270	0.306	0.342	0.367	0.384	0.395	0.463	0.411	0.358	0.325	0.266	0.246	
	taud	1.924	1.973	1.978	2.044	2.254	2.362	2.131	2.274	2.387	2.240	1.981	1.879	
	Ebn at Noon	400	629	745	812	835	834	753	769	752	626	404	256	
	Edh at Noon	47	85	119	136	120	110	136	109	81	65	43	30	
All-Sky Solar Radiation	RadAvg	0.39	1.27	2.66	4.20	4.92	5.54	5.11	3.62	2.15	0.93	0.43	0.19	
	RadStd	0.05	0.10	0.27	0.46	0.55	0.60	0.44	0.33	0.29	0.14	0.09	0.03	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (7 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page